

INNOPOLIS UNIVERSITY · BACHELOR'S THESIS · 2026

Secure Communication for Swarm Robotics

Adaptive lightweight cryptography that switches at runtime.

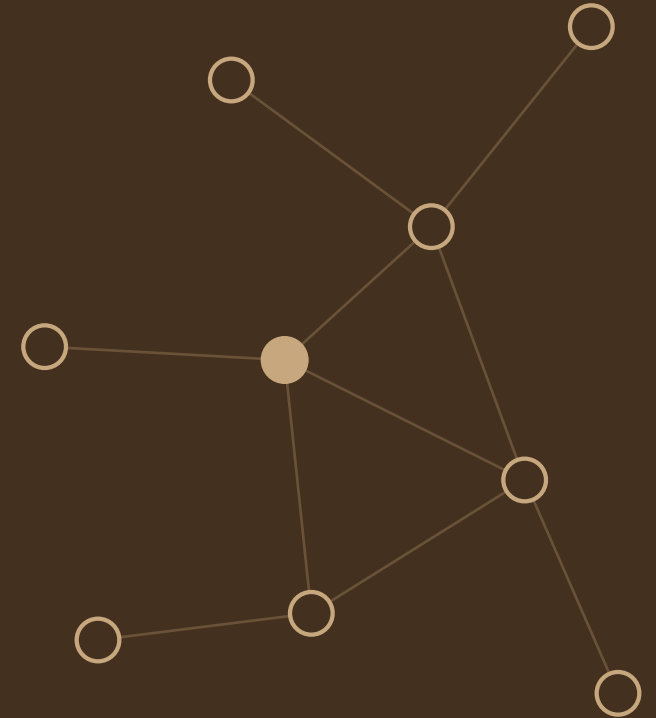
search & rescue · agriculture · inspection · defence

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SUPERVISOR

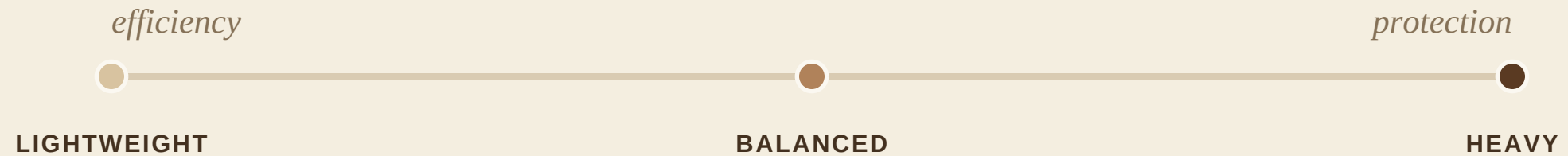
Oleg Bulichev



THE PROBLEM

No single cipher fits a whole mission

Security and performance are coupled —
so balance them while the mission runs.



Always heavy

wastes throughput & battery

Always light

unsafe when threat rises

Strong primitives — but fixed for the whole mission

— WHAT THE FIELD OFFERS

- **Lightweight crypto**
ECC, AEAD, hash tokens — efficient on constrained nodes
- **Decentralised trust**
blockchain & attestation — strong assurance, heavy stack
- **Resilient comms**
self-healing relays, risk-aware routing — little built-in security

— THE GAP

- **Static configuration**
security parameters fixed for the whole mission
- **Fragmented layers**
trust, resilience & message protection studied apart
- **No executable comparison**
adaptive ideas rarely benchmarked against static baselines

Adaptive control of crypto settings is studied far less than the primitives themselves.

AIM & RESEARCH QUESTION

Does adapting crypto at runtime pay off?

— *can it hold the right protection at lower cost than a fixed profile?*

HOW THIS THESIS ANSWERS IT

01

MODEL

Swarm graph + adaptation function

$F : S(t) \rightarrow P$

02

PROTOTYPE

Pairwise testbed, four modes

incl. adaptive switching

03

EVIDENCE

Measured in running code

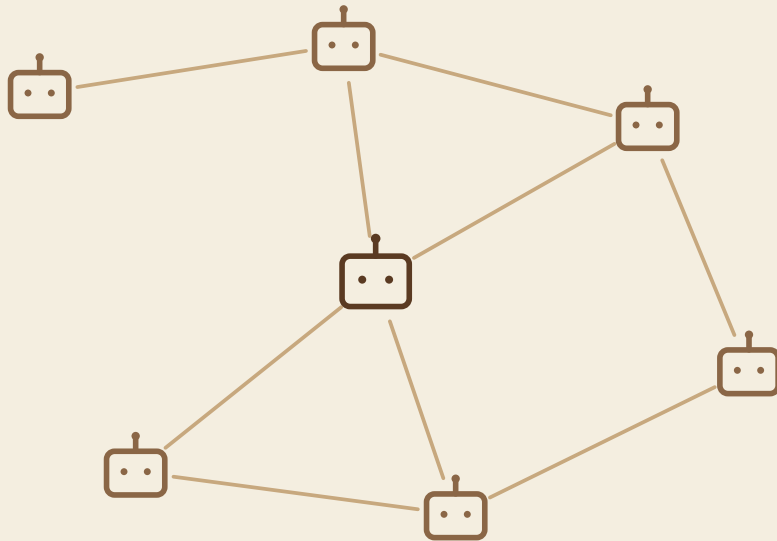
228 benchmark runs

Novelty: a bounded formal model paired with an executable benchmark.

The swarm as a dynamic graph

$$G (V, E, t)$$

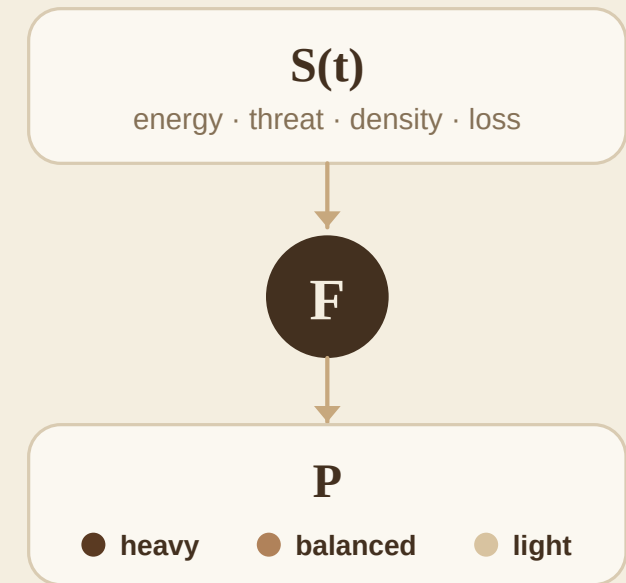
nodes = agents · edges = live links



each node carries energy · keys · position

— THE ADAPTATION FUNCTION

$$F : S(t) \rightarrow P$$



Three threats, and a data plane that adapts

— THE ADVERSARY CAN



Eavesdrop
read traffic



Tamper
inject / modify



Compromise
leak keys & memory

— OUR DEFENCE

SESSION SETUP

X25519 + HKDF-SHA256

forward secrecy · epoch rekeying

fixed



DATA PLANE

switches per tick

- **Heavy** AES-256-GCM + HMAC
- **Balanced** AES-192-GCM + HMAC
- **Lightweight** ChaCha20-Poly1305 + token

PROFILES

Three operating points

HEAVY

AES-256-GCM

+ HMAC-SHA256

0.250 MiB

crypto work / message

BALANCED

AES-192-GCM

+ HMAC-SHA256

0.250 MiB

crypto work / message

LIGHTWEIGHT

ChaCha20-Poly1305

+ hash token

0.125 MiB

crypto work / message

Lightweight does half the per-message crypto work of the authenticated profiles.

This metric counts bytes processed, so heavy and balanced come out equal; they differ in key length.

BENCHMARK

A constrained, reproducible testbed



two containers · encrypted data path

0.05

CPU CORES

128

MiB RAM

64

KiB / MSG

60

SECONDS

×3

REPEATS

Measured: latency · throughput · drops · switches · crypto work

Adaptive avoids both failure modes

Every static profile fails one way; adaptive fails neither.

PROFILE	UNDER-PROTECTS	OVER-PROTECTS	PROTECTION COST vs. always-heavy
Always-heavy	never	60% of run	100%
Always-balanced	40% of run	20% of run	67%
Always-light	80% of run	never	33%
Adaptive	never	never	73%

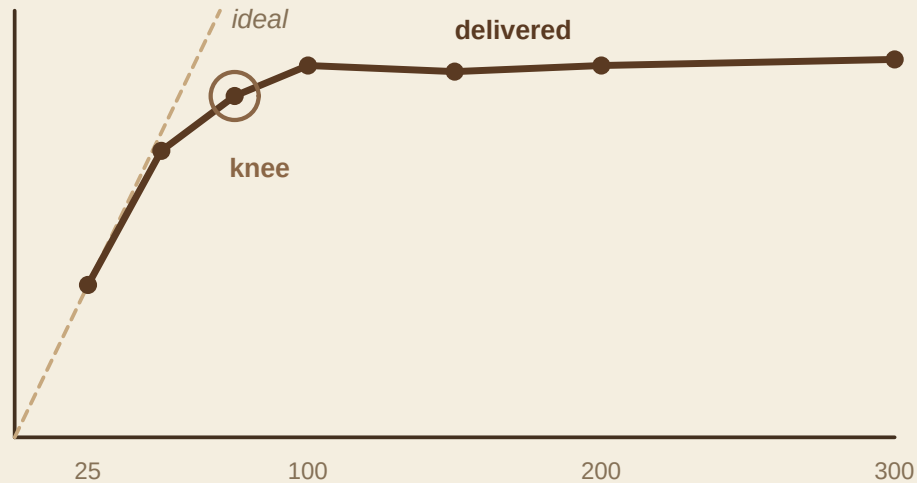
Only adaptive is never wrong either way — at 27% less protection work than always-heavy.

And the cost tracks the mission — a calmer schedule trends toward lightweight's cost, still escalating when it matters.

Under a tight CPU cap, transport dominates

DELIVERED vs OFFERED RATE

0.05-core cap · msg/s



CPU CAP REMOVES THE BOTTLENECK



The ceiling is CPU.

The crypto design keeps up.

Near saturation the pipeline converges; the profile difference shows up in crypto work and policy fit.

CONCLUSION

Adaptive crypto, built and measured

✓ Formalised

swarm graph + $F : S(t) \rightarrow P$

✓ Built

four modes incl. adaptive switching

✓ Measured

lightweight cheapest; adaptive allocates well

WHAT COMES NEXT

01 Authenticated control plane

signed bootstrap & updates

02 Multi-peer validation

churn · contention · async

03 Hardware & post-quantum

real platforms; hybrid PQ

Thank you — questions welcome.

github.com/mazzz3r/adaptive-swarm-crypto-thesis

Limitations & scope

- **Pairwise prototype**

two peers on the measured path; multi-hop & churn are out of scope

- **Energy is a model proxy**

valid for relative comparison only, not battery-accurate

- **Data plane only**

no authenticated control plane (signed bootstrap / updates)

- **Host environment**

Docker + gRPC containers; no embedded radios or microcontrollers

- **Scripted inputs**

threat & energy are scripted, not sensed or estimated

- **Coarse cost metric**

crypto-work counts bytes, so heavy $\approx$ balanced; thresholds set by hand

Every claim covers exactly what the constrained pairwise prototype actually runs.